

When Can Row Scaling Make Integer Rounding Safe?

Working research note: geometry, certificates and local strengthening

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This note develops a possible replacement for the previous paper’s central contribution. It is not a claim of established publication novelty. Farkas separation, nonnegative aggregation, safe cut rounding, Boolean no-good clauses and geometric programming are established tools. The question is whether a useful, sufficiently distinct result arises from characterizing which numerical acceptance failures row scaling can repair.

1 A universal acceptance property

Let $P = \{x : Ax \leq b, 0 \leq x \leq 1\}$, with rational data and n binary variables. For a fixed $0 \leq \eta < 1/2$, the rounding cell of $z \in \{0, 1\}^n$ is

$$C_z = z + E_z, \quad E_z = \{e : -\eta \mathbf{1} \leq e \leq \eta \mathbf{1}, 0 \leq z + e \leq 1\}.$$

The theorems below use these exact bounds. The implementation also certifies acceptance with an enlarged bound-tolerance box, while the intrinsic obstruction test retains the original exact bounds. A bad center is an integer $z \notin P$. A nonnegative vector t specifies an original-row residual hypothesis: $A(z + e) \leq b + t$. For exported row tolerances $\epsilon_r > 0$ and positive row factors d_r , $t_r = \epsilon_r/d_r$. Numerical safety means that every point satisfying this hypothesis and the rounding-cell condition rounds to a feasible integer center. It is a universal conditional property, not a statement that a solver enforces its input tolerances on the exported rows or proves a correct optimal bound.

Theorem 1 (Exact cell exclusion). *A bad center z cannot be accepted if and only if there exists $\lambda \geq 0$ such that*

$$\lambda^T(Az - b - t) > h_{E_z}(-A^T\lambda), \quad h_E(v) = \max_{e \in E} v^T e. \quad (1)$$

Proof. Acceptance is feasibility of $Ae \leq b + t - Az$ in the compact nonempty box E_z . If (1) holds, multiplying those inequalities by λ contradicts $\lambda^T Ae \geq -h_{E_z}(-A^T\lambda)$. Conversely, infeasibility separates the compact convex set AE_z from the downward orthant translated by $b + t - Az$. The separating normal is nonnegative because that orthant is unbounded downward; strict separation gives the displayed inequality. $\square$

For a symmetric error box, the support is $\sum_j \eta_j |(A^T\lambda)_j|$. Cancellation occurs before the absolute values. A row-by-row certificate restricts λ to unit vectors; matching the actual one-sided binary bounds alone does not recover all multirow certificates.

Proposition 2 (Finite complete family for two rows). *For a two-row block, it suffices to test $\lambda = (s, 1 - s)$ at $s = 0$, $s = 1$, and all values in $(0, 1)$ satisfying $sa_{1j} + (1 - s)a_{2j} = 0$ for a column whose expression is not identically zero. Zero columns are ignored. There are at most $n + 2$ candidates. Thus every binary rounding cell can be tested using rational arithmetic without an optimizer; all cells can be tested in $O(2^n n^2)$ rational arithmetic operations.*

Proof. Normalize a nonzero separating multiplier by its positive coordinate sum. The certificate margin is a continuous concave piecewise-affine function of s . Each box support term changes its affine expression only when the corresponding coordinate of $A^T \lambda$ changes sign. An affine function on each closed interval between consecutive breakpoints attains its maximum at an endpoint. Testing all breakpoints and endpoints is therefore complete. Evaluating a candidate costs $O(n)$. The operation count does not suppress the bit cost of rational arithmetic. $\square$

Proposition 3 (Complete finite family for a fixed number of rows). *Let $m \geq 1$. Intersect the multiplier simplex $\{\lambda \geq 0 : \mathbf{1}^T \lambda = 1\}$ with the column hyperplanes $(A^T \lambda)_j = 0$. The vertices of the resulting closed sign cells form a complete certificate family $V(A)$, independently of z, b, t and the box endpoints. It suffices to consider at most $\binom{n+m}{m-1}$ candidates. Every required ray has support at most $\min\{m, n + 1\}$.*

Proof. On each closed sign cell the box support, and hence the certificate margin, is affine. Its maximum is attained at a vertex. A vertex of the simplex has $m - 1$ independent active equations chosen from its coordinate facets and the column hyperplanes; solving these equations with $\mathbf{1}^T \lambda = 1$ enumerates all candidates. Singular systems are skipped. If a vertex has support s , its relative simplex has dimension $s - 1$, requiring $s - 1$ independent column cancellations. Thus $s \leq n + 1$. Equivalently, enumerate each support of size s and each set of $s - 1$ columns. The number of systems is bounded by

$$\sum_{s=1}^{\min(m, n+1)} \binom{m}{s} \binom{n}{s-1} = \binom{n+m}{m-1}.$$

Duplicate vertices and degenerate hyperplanes only reduce the output size. $\square$

This is an application of classical hyperplane-arrangement and zonotope-normal geometry, not a new facet-enumeration construction. For fixed m the multiplier enumeration is polynomial in n , but testing every binary center still costs 2^n times that amount. The implementation uses exact rational elimination; its arithmetic and bit costs must be distinguished.

Proposition 4 (Exact weighted residual threshold). *Fix positive weights $w \in \mathbb{Q}_{>0}^m$. For a bad center define*

$$R_z = \max_{\lambda \in V(A)} \frac{\lambda^T (Az - b) - h_{E_z}(-A^T \lambda)}{\lambda^T w}.$$

Then

$$R_z = \min_{e \in E_z} \max_r \frac{a_r^T(z + e) - b_r}{w_r}.$$

If there is at least one bad center, put $R = \min_{z \text{ bad}} R_z$. For $s \geq 0$, the acceptance contract $Ax \leq b + sw$ is safe if and only if $s < R$. Equality is unsafe. If there are no bad centers, every $s \geq 0$ is safe.

Proof. Normalize multipliers by $\lambda^T w = 1$ and apply the minimax theorem to the bilinear expression $\lambda^T (A(z + e) - b)$ on two compact convex sets. Its maximum over normalized multipliers at a fixed e is the displayed maximum row residual. On a sign cell of the usual unit-sum simplex, the numerator

is affine and the denominator is strictly positive affine. Its ratio at a convex combination of vertices is a convex combination of their ratios, with weights proportional to the positive denominators. A maximizing vertex therefore suffices. Finally, compactness makes $s \geq R_z$ equivalent to the existence of an accepted point in that cell. Taking the minimum over bad centers proves the strict threshold. $\square$

For fixed positive reference scales d_r^0 and exported tolerances ϵ_r , take $w_r = \epsilon_r/d_r^0$. A common extra factor α is safe precisely when $1/\alpha < R$. If $R > 0$, this gives the sharp condition $\alpha > 1/R$; if $R \leq 0$, no such factor works. This quantifies the earlier qualitative obstruction. The result is conditional on the exported inequalities and uses exact stored data.

A strict improvement over bound-aware individual rows. Take $M = 10^9$, $\eta = 10^{-6}$,

$$A = \begin{bmatrix} M & 1-M \\ 1-M & M \end{bmatrix}, \quad b = (1/2, 1/2)^T, \quad t = (1/4, 1/4)^T.$$

Only $(0,0)$ is integer feasible. At $z = (1,1)$ each individual row can hide its violation within the one-sided cell: decreasing the coordinate with positive coefficient by η changes its activity by $M\eta$. But the combined row $x_1 + x_2 \leq 1$ gives a certificate whenever $t_1 + t_2 < 1 - 2\eta$. The other bad centers $(1,0)$ and $(0,1)$ are excluded by individual rows. The joint certificate therefore establishes a property that no collection of independent row certificates establishes. This example remains valid with a bound tolerance of η , using the enlarged symmetric error box.

2 What row scaling cannot repair

Theorem 5 (Intrinsic obstruction and scaling remediability). *Fix the rounding cells. There exists some finite positive diagonal row scaling that makes the exported-residual acceptance property safe if and only if $P \cap C_z = \emptyset$ for every bad center z . If an intersection is nonempty, no positive row scaling and no collection of inequalities valid for the whole LP relaxation P can remove that obstruction.*

Proof. An exact feasible point in a bad rounding cell stays feasible under positive scaling and under every LP-valid added row. Conversely, there are finitely many bad centers. For each, compactness gives a strictly positive minimum

$$\rho_z = \min_{e \in E_z} \max_r \{a_r^T(z + e) - b_r\}$$

when the cell does not meet P . Their minimum ρ is positive. A common factor $d > \max_r \epsilon_r/\rho$ excludes all bad cells under the exported-row hypothesis. If there are no bad centers, safety is immediate. $\square$

With coefficient ceilings, each row has a maximum admissible factor H_r . Acceptance sets shrink monotonically as factors increase, so a feasible scaling exists within the factor box if and only if its componentwise maximum H is safe. This test can be costly even though monotonicity is elementary.

Theorem 6 (Complexity of the intrinsic safety test). *For any fixed rational $0 < \eta < 1/4$, deciding whether every point of P within η of a binary center rounds to a feasible center is coNP-complete, even when P has a known binary feasible point and all matrix entries belong to $\{-1, 0, 1\}$.*

Proof. Reduce a 3-SAT instance with distinct variables per clause. Introduce its binary variables x and another binary variable y . For every clause with literal functions ℓ_1, ℓ_2, ℓ_3 , impose

$$y \leq \ell_1(x) + \ell_2(x) + \ell_3(x), \quad y \leq 1 - \eta.$$

All variables have bounds $[0, 1]$. The matrix has entries in $\{-1, 0, 1\}$; $(x, y) = (0, 0)$ is an integer feasible point. Every binary point with $y = 0$ is feasible. If the formula is satisfiable, its satisfying assignment with $y = 1 - \eta$ belongs to P and rounds to an infeasible center with $y = 1$. Conversely, an intrinsic bad cell must have rounded $y = 1$. If its rounded x violates a clause, each of that clause's three literal values at the nearby continuous point is at most η , whereas $y \geq 1 - \eta$. This contradicts the clause row because $1 - \eta > 3\eta$. Thus an intrinsic bad cell implies a satisfying assignment. The complement is in NP: a bad binary center together with a rational feasible point in its cell is a polynomially verifiable witness. A nonempty bounded rational polytope has a rational vertex of polynomial encoding length. $\square$

The construction applies to a fixed tolerance such as 10^{-6} . The difficulty is not caused by large coefficients. It also distinguishes a certificate of safety for all possible returned points from checking one already returned solution.

Corollary 7 (Sparse unit-coefficient hardness). *The preceding hardness result still holds when every row of A has at most four nonzeros and every column of A has at most five nonzeros.*

Proof. Give every occurrence of each SAT variable its own copy and link successive copies by equalities. Likewise use one gate variable y_c per clause, link the gates in a chain, and impose the cap $y_1 \leq 1 - \eta$ at an endpoint. Each equality is written as two inequalities. Exact equality forces equal rounded centers since $\eta < 1/2$. Consequently a bad cell cannot arise from inconsistent copies: its gate centers must all equal one, and the earlier satisfying-assignment argument applies. The all-zero point remains integer feasible. A copy occurs in one clause and at most two chain equalities, for at most five inequality rows. The capped gate is at an endpoint, so it occurs in at most four rows. Clause rows contain four variables, equality rows two, and the cap one. The construction is polynomial. $\square$

A limitation of this hardness construction. The theorem concerns the given LP relaxation. Integer bound tightening immediately sets $y = 0$ in the constructed instance and removes the obstruction. It therefore does not establish that finding an integer-equivalent safe reformulation is hard. Indeed, for integer rows with integer right-hand sides, the familiar rowwise lattice test $\epsilon + \eta \|a_r\|_1 < 1$ already guarantees rounded feasibility. Fractional right-hand sides near an integer are essential to this unit-coefficient construction.

Proposition 8 (One-row tractability at a fixed tolerance). *For one rational inequality in binary variables and a fixed rational $0 < \eta < 1/2$, intrinsic safety is decidable in polynomial time in the number of variables and the data encoding length. An explicit algorithm uses $O(n^{\lceil 1/\eta \rceil})$ rational arithmetic operations. The exponent makes this a theoretical tractability result, not a practical bound at solver tolerances. If η is part of the input, the one-row problem is coNP-complete, even for positive integer coefficients, an integer right-hand side and a known feasible binary point.*

Proof. Complement variables with negative coefficients, obtaining $\sum_j a_j x_j \leq B$ with $a_j \geq 0$. Complementation preserves binary rounding distances. Cases $B \leq 0$ and $B \geq \sum_j a_j$ are immediately safe: the relaxation is empty, permits only zero-weight coordinates, or contains every binary center, respectively. Assume $0 < B < \sum_j a_j$.

A binary subset with weight S has a bad cell meeting the relaxation precisely when

$$B < S \leq U := \frac{B}{1 - \eta}.$$

The minimum activity in its cell is $(1 - \eta)S$. Put $\Delta = U - B > 0$ and ignore items heavier than U . Call weights above Δ large and positive weights at most Δ small. A witness contains strictly fewer than $U/\Delta = 1/\eta$ large items. Enumerate every possible set of at most $\lceil 1/\eta \rceil - 1$ large items, with total L . If $B < L \leq U$, a witness has been found. If $L \leq B$ and adding all small items can exceed B , add small items until the first strict crossing. The total is then at most $B + \Delta = U$. These cases are exhaustive: any witness's large-item set is enumerated and its small-item subset cannot outweigh the set of all small items. Enumeration has the stated polynomial bound for fixed η . The produced near-binary witness is $x = (1 - \eta)z$ in the complemented coordinates.

For hardness when η is input, take an instance of SUBSET SUM with positive integer weights and integer target $T \geq 3$. Set $B = T - 1$ and $\eta = 1/T$. Then $U = T$, so an integer subset weight in $(B, U]$ is exactly a subset of weight T . The origin is feasible, the reduction has polynomial encoding length, and a witness subset is polynomially verifiable. Taking complements gives the result. $\square$

This proof specializes an interval subset-sum argument; it makes no claim of a new knapsack algorithm. It complements the multirow hardness result by identifying which parameters cause the obstruction-recognition problem to be hard. Exact margin computation remains harder than testing its sign: for one positive row, the threshold in Proposition 4 with unit weight equals $(1 - \eta)S^* - B$, where S^* is the minimum attainable subset weight strictly above B . Computing that exact minimum contains SUBSET SUM even for fixed η .

3 From diagnosis to an explicit repair

For a small generator block, enumerate its integer centers and classify each bad one under the intended scaled-row acceptance model. Three actions are distinct:

1. A current joint certificate already excludes the cell; no new row is needed.
2. An LP-valid aggregate excludes the cell after suitable numerical normalization; adding a redundant row can enforce a useful acceptance margin.
3. The original LP meets the bad cell; an LP-redundant row cannot repair it. An integer-valid strengthening or a tighter integrality requirement is needed.

For the third case, our exploratory implementation uses established Boolean no-good clauses. A partial assignment $(z_j)_{j \in S}$ occurring in no local feasible integer center yields

$$\sum_{j \in S: z_j=1} x_j - \sum_{j \in S: z_j=0} x_j \leq |\{j \in S : z_j = 1\}| - 1.$$

It excludes any matching binary center by one unit; the margin remains at least $1 - |S|\eta$ in its rounding cell. Small integer coefficients make a positive exported-residual tolerance easy to accommodate. Selecting clauses only for cells not already excluded is a finite coverage problem. The prototype uses deterministic greedy coverage with sparsity tie-breaking; it does not claim minimum cardinality. These are familiar cuts, and their selection must be compared with full local strengthening and ordinary presolve.

Preserving LP validity when an aggregate is written in binary64. Let the exact nonnegative aggregate be $p^T x \leq q$ on $[0, 1]^n$, and let $\hat{p}$ be its rounded coefficient vector after a positive normalization. Writing

$$\hat{q} = \text{roundUp} \left(q + \sum_j \max\{\hat{p}_j - p_j, 0\} \right)$$

preserves validity for every point of P . The correction is the exact box support of the coefficient-rounding error. A subsequent cell-exclusion check must use the actually exported $(\hat{p}, \hat{q})$, so outward relaxation is paid for in the certificate. This is safe rounding of an established aggregation, not a novel cut family.

Scope of the local-to-global claim. If every original local row belongs to a block whose rounded centers are certified, then rounding preserves those local rows. Coupling rows need their own certificates; local safety does not imply global safety. In the experiment, the single coupling row has integer coefficients and integer RHS, and its unit lattice violation exceeds $\epsilon + \eta \|a\|_1$. This certifies that row separately. General mixed continuous coupling requires an additional argument and is not covered by this experiment.

4 Relation to prior work and research gates

The numerical acceptance model is already discussed in the MIPLIB 2010 paper, <https://doi.org/10.1007/s12532-011-0025-9>. Feasible rounding through inner parallel sets and granularity is developed by Neumann, Stein and Sudermann-Merx; see <https://doi.org/10.1007/s10013-020-00393-4> and its references. Coey, Lubin and Vielma explicitly scale conic certificate cuts to maintain guarantees under LP feasibility tolerances, <https://doi.org/10.1007/s12532-020-00178-3>, Section 3.2. Consequently, certificate-based tolerance scaling alone is not a defensible novelty claim. Standard presolve reductions, including GCD and RHS strengthening, are reviewed at <https://doi.org/10.1287/ijoc.2018.0857>.

Finite multiplier enumeration also uses established zonotope facet-normal geometry; see Althoff, Stursberg and Buss, Theorem 7, <https://mediatum.ub.tum.de/doc/1287515/document.pdf>. Local table encodings and strengthened table cuts for ILP are studied by Bierlee, Piessens, Guns and Stuckey (CP 2026), <https://doi.org/10.4230/LIPIcs.CP.2026.6>; automatic subproblem tabulation is studied by Akgün et al. (JAIR 2025), <https://doi.org/10.1613/jair.1.17032>. These results further restrict any compiler novelty claim. Selecting forbidden partial patterns for unsafe centers, while treating already-excluded centers as don't-cares, is a Boolean minimization problem. The numerical construction of that care set is the question under study.

Potentially distinct content is the exact row-scaling remediability characterization, the complete finite two-row certificate family, and a compiler that separates LP-redundant repairs from necessary integer-domain strengthening before choosing sparse additions. Establishing publication merit requires a careful prior-art audit, independent verification of the hardness argument and certificates, and evidence that this distinction is useful beyond constructed examples.

The current experiments retain exact feasible-rounded-point and optimality outcomes separately. A local feasibility certificate cannot justify a solver's global dual bound. The early pilot already contains feasible but suboptimal outputs labeled optimal by HiGHS, making that limitation operationally relevant. No speedup or solver-independent optimality guarantee is asserted.

5 A completed control experiment and its negative conclusion

The revised synthetic study uses twelve 72-variable binary models: four independent random seeds, each at three magnitudes $M \in \{10^6, 10^9, 10^{12}\}$. Each model has twelve six-variable blocks and an integer resource constraint. Within a block,

$$A_k = \begin{bmatrix} Mv_k^T + u_k^T \\ -Mv_k^T + w_k^T \end{bmatrix}, \quad v_k \in \mathbb{Z}^6, \quad u_k, w_k \in \{1, \dots, 7\}^6.$$

Right-hand sides are half-integers of moderate magnitude. The original integer optimum is computed by dynamic programming over the exact local feasible states, then independently recomputed. Eight original methods produce 384 assigned configurations with two solvers and presolve on/off. An additional classical control adds 48 configurations. Forty-eight assignments abstain without calling a solver; the complete comparison therefore contains 384 actual solver calls. Native limits are ten seconds, with an external 25-second limit and one thread. Failures and timeouts remain in the denominator. The magnitude and presolve repetitions are not independent statistical replicates.

An exact obstruction, independent of solver behavior. An independent rational vertex-enumeration routine constructs 83 exact feasible points in bad rounding cells, spread across all twelve original models. Each local witness is embedded into the full model by setting the other blocks to zero; every original row, including the resource row, is checked exactly. These are explicit obstructions to safety under *any* row scaling or LP-valid added rows, regardless of their numerical normalization. They are not inferred from solver failures.

A necessary classical reformulation control. Because $v_k^T z$ is integral and M dominates the smaller terms and right-hand sides, an integer feasible point necessarily satisfies $v_k^T z = 0$. Thus the original local integer set is exactly represented by

$$v_k^T z = 0, \quad u_k^T z \leq \lfloor b_{k1} \rfloor, \quad w_k^T z \leq \lfloor b_{k2} \rfloor.$$

For example, $v_k^T z \geq 1$ implies $Mv_k^T z + u_k^T z \geq M + \sum_j \min(u_{kj}, 0) > b_{k1}$; the other sign is analogous. The experiment checks these strict dominance inequalities and verifies equality of the two integer sets on all 64 local centers. This is a generator-aware lattice reformulation using classical reasoning, not a new algorithm.

Method	Gurobi / 24		HiGHS / 24		Median rows
	Feasible	Optimal	Feasible	Optimal	
Original	12	7	15	15	25
GCD and RHS flooring	15	10	15	15	25
Targeted integer clauses	24	24	24	18	37
Full clause cover	24	24	24	20	125.5
Lattice decomposition	24	24	24	24	49

Table 1: Exact checks of rounded outputs against the original integer model and its independently computed optimum. Twenty-four configurations per solver comprise twelve models and two presolve settings. Full results include the remaining four controls and all abstentions.

The targeted clauses certify all local bad-cell exclusions under the declared exported-row and rounding hypotheses, using 145 clauses across 144 blocks. They return feasible rounded outputs in

all 48 configurations, but six HiGHS outputs marked optimal are suboptimal; all six use presolve. Full clause covers also recover feasibility but retain four suboptimal HiGHS outputs. Keeping the original ill-scaled rows therefore leaves numerical optimality problems that the primal safety certificate does not address.

The lattice reformulation returns an exactly optimal rounded point in all 48 configurations. Its median preparation time is 0.00045 seconds, compared with 0.313 seconds for targeted clauses and 0.050 seconds for the full clause cover. These small local timings are descriptive, exclude independent post-run validation, and do not establish a general speedup. They are sufficient to reject a claim that the targeted compiler is preferable on this family merely because it adds fewer rows. Further expansion of this same synthetic family cannot substitute for a different application or a sufficiently substantive theoretical contribution.

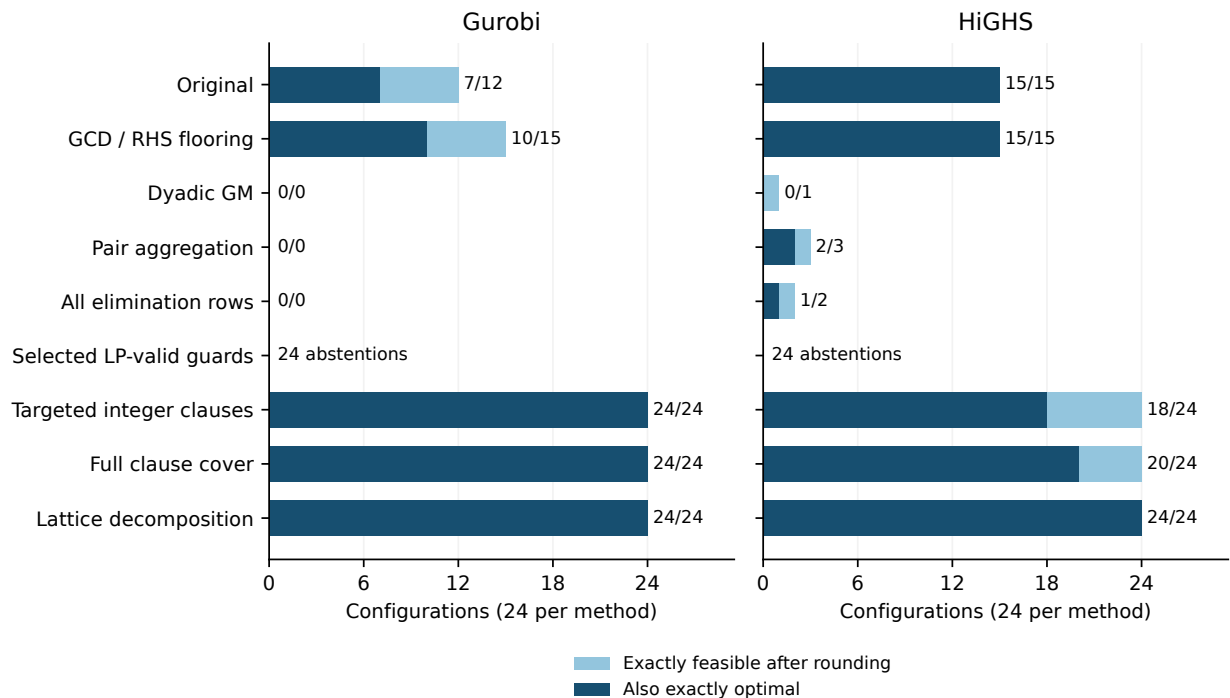


Figure 1: All nine methods, including the late-added lattice control. Light bars count exact feasible rounded points and dark bars count exact optima. Numeric labels give optimal/feasible counts, each out of 24; they are not success-rate fractions. The selected LP-valid guard method abstains on every assigned model.

Reproducibility of this investigation. Saved inputs, source snapshots, assignments, outputs and exact certificates are in `results-revision/research-audit/guards-holdout-100-isolated/` and `results-revision/research-audit/guards-lattice-control-100/`. Separate saved-data and rational-witness checkers do not import the certificate compiler or an optimizer. The expanded mathematical routines are covered by independent LP comparisons, exhaustive exact one-row checks, reduction examples and boundary tests. The main submission package predates these results and remains a superseded candidate.